

CSS text-orientation Property
Complete Guide to Text Direction and Vertical Writing

CSS Typography Mastery

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Chapter 1

Introduction to text-orientation

1.1 What is text-orientation?

The `text-orientation` CSS property sets the orientation of text characters in writing modes that support vertical text. It works in conjunction with `writing-mode` to control how text is displayed vertically or horizontally.

Key facts about text-orientation:

- Controls character orientation in vertical writing systems
- Works with `writing-mode` property
- Supports vertical-rl, vertical-lr, and sideways modes
- Values: mixed, upright, sideways, use-glyph-orientation
- Essential for CJK (Chinese, Japanese, Korean) text
- Part of CSS Writing Modes Module Level 3
- Widely supported in modern browsers
- Affects text layout and readability

1.2 Why Use text-orientation?

Text orientation is crucial for international typography:

- **CJK support:** Enable proper display of Chinese, Japanese, Korean
- **Vertical writing:** Support traditional vertical text layouts
- **International reach:** Make content accessible globally
- **Design flexibility:** Create unique layouts with vertical text
- **Typography control:** Fine-tune character display
- **Cultural authenticity:** Respect traditional writing systems
- **Improved readability:** Proper orientation for vertical text

1.3 Browser Compatibility

Text-orientation is supported in modern browsers:

- Chrome 48+ (2015)
- Firefox 41+ (2015)
- Safari 11+ (2017)
- Edge 79+ (2020)
- Opera 35+ (2015)
- Mobile browsers (most modern versions)

Chapter 2

Basic Syntax and Values

2.1 text-orientation Syntax

```
/* Basic syntax */
text-orientation: mixed | upright | sideways | use-glyph-orientation;

/* With writing-mode */
.vertical-text {
    writing-mode: vertical-rl;
    text-orientation: mixed;
}

/* Upright orientation */
.upright {
    writing-mode: vertical-lr;
    text-orientation: upright;
}

/* Sideways orientation */
.sideways {
    writing-mode: vertical-rl;
    text-orientation: sideways;
}

/* Global values */
text-orientation: inherit;
text-orientation: initial;
text-orientation: revert;
```

2.2 text-orientation Values

Value	Description
<code>mixed</code>	Default. Characters rotate 90 degrees, Latin text upright
<code>upright</code>	All characters rotated to upright (0 degrees)
<code>sideways</code>	All characters rotated 90 degrees clockwise
<code>use-glyph-orientation</code>	Use font's natural glyph orientation

2.3 Understanding text-orientation Values

```
/* mixed - default, most common for CJK */
.text-mixed {
    writing-mode: vertical-rl;
    text-orientation: mixed;
    /* CJK characters vertical, Latin horizontal rotated */
}

/* upright - all characters upright */
.text-upright {
    writing-mode: vertical-rl;
    text-orientation: upright;
    /* All characters appear upright */
}

/* sideways - all characters rotated 90 degrees */
.text-sideways {
    writing-mode: vertical-rl;
    text-orientation: sideways;
    /* All characters rotated like horizontal text */
}

/* use-glyph-orientation - uses font metrics */
.text-glyph {
    writing-mode: vertical-rl;
    text-orientation: use-glyph-orientation;
    /* Depends on font */
}
```

Chapter 3

Working with writing-mode

3.1 text-orientation and writing-mode

```
/* Vertical writing modes */

/* Vertical right-to-left (most common) */
.vertical-rl {
    writing-mode: vertical-rl;
    text-orientation: mixed;
}

/* Vertical left-to-right */
.vertical-lr {
    writing-mode: vertical-lr;
    text-orientation: mixed;
}

/* Sideways writing mode */
.sideways-rl {
    writing-mode: sideways-rl;
    text-orientation: upright;
}

/* Horizontal writing (default) */
.horizontal {
    writing-mode: horizontal-tb;
    text-orientation: mixed;
}
```

3.2 Writing Mode Combinations

```
<style>
/* Japanese vertical text */
.japanese-vertical {
    writing-mode: vertical-rl;
    text-orientation: mixed;
    font-size: 1.2rem;
}
```

```
    line-height: 2;
  }

  /* Chinese vertical text */
  .chinese-vertical {
    writing-mode: vertical-lr;
    text-orientation: upright;
    font-family: 'Noto Sans CJK SC', sans-serif;
  }

  /* Korean vertical (less common but supported) */
  .korean-vertical {
    writing-mode: vertical-rl;
    text-orientation: mixed;
  }

  /* Sideways text (for labels, titles) */
  .sideways-text {
    writing-mode: sideways-rl;
    text-orientation: sideways;
  }
</style>

<div class="japanese-vertical">

</div>

<div class="chinese-vertical">

</div>

<div class="sideways-text">
  Vertical Label
</div>
```


Chapter 4

Practical Examples

4.1 Japanese Vertical Text

```
<style>
  body {
    font-family: 'Noto Sans JP', 'Hiragino Kaku Gothic Pro',
      sans-serif;
  }

  .japanese-article {
    writing-mode: vertical-rl;
    text-orientation: mixed;
    max-height: 600px;
    columns: 2;
    column-gap: 2rem;
    padding: 2rem;
    background: white;
    border: 1px solid #ddd;
    border-radius: 8px;
  }

  .japanese-article h1 {
    writing-mode: vertical-rl;
    text-orientation: mixed;
    text-align: center;
    margin-bottom: 2rem;
    font-size: 1.5rem;
  }

  .japanese-article p {
    margin: 0;
    line-height: 1.8;
  }

  /* Ruby annotations for furigana */
  ruby {
    font-size: 0.75em;
  }
```

```
    rt {  
        writing-mode: horizontal-tb;  
    }  
</style>  
  
<article class="japanese-article">  
    <h1>    </h1>  
    <p>  
  
    </p>  
</article>
```

4.2 Chinese Vertical Text

```
<style>  
    .chinese-article {  
        writing-mode: vertical-lr;  
        text-orientation: upright;  
        font-family: 'Noto Sans CJK SC', SimSun, sans-serif;  
        max-height: 500px;  
        columns: 3;  
        column-gap: 2rem;  
        padding: 2rem;  
        background: #f9f9f9;  
        border-radius: 8px;  
    }  
  
    .chinese-article h1 {  
        writing-mode: vertical-lr;  
        text-orientation: upright;  
        text-align: center;  
        margin-bottom: 1.5rem;  
        font-size: 1.4rem;  
        font-weight: 600;  
    }  
  
    .chinese-article p {  
        margin: 0;  
        text-align: justify;  
        line-height: 2;  
    }  
  
    .traditional-format {  
        writing-mode: vertical-rl;  
        text-orientation: mixed;  
    }  
</style>  
  
<article class="chinese-article">
```

```
<h1> </h1>
<p>

</p>
</article>
```

4.3 Mixed Language Vertical Text

```
<style>
  .mixed-language {
    writing-mode: vertical-rl;
    text-orientation: mixed;
    font-family: 'Noto Sans JP', Arial, sans-serif;
    padding: 1.5rem;
    background: white;
    border: 1px solid #ddd;
    line-height: 2;
  }

  .mixed-language span.english {
    /* English text rotates in mixed mode */
    display: inline-block;
    transform: rotate(0deg);
  }

  .mixed-language span.japanese {
    /* Japanese text stays vertical in mixed mode */
    display: inline;
  }
</style>

<div class="mixed-language">
  <p>
    <span class="japanese"> </span>
    <span class="english">English</span>
    <span class="japanese"> </span>
    <span class="english">Japanese</span>
    <span class="japanese"> </span>
  </p>
</div>
```

4.4 Sideways Text Labels

```
<style>
  /* Chart axis labels */
  .chart-label {
```

```
        writing-mode: sideways-rl;
        text-orientation: sideways;
        font-weight: 600;
        padding: 0.5rem;
    }

    /* Table column headers */
    .table-header-vertical {
        writing-mode: vertical-rl;
        text-orientation: upright;
        padding: 1rem;
        background: #f5f5f5;
        border: 1px solid #ddd;
    }

    /* Book spine text */
    .book-spine {
        writing-mode: vertical-rl;
        text-orientation: mixed;
        padding: 1rem;
        background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);
        color: white;
        font-weight: 600;
    }
</style>

<!-- Chart axis label -->
<div class="chart-label">Sales (2024)</div>

<!-- Table header -->
<table>
    <thead>
        <tr>
            <th class="table-header-vertical">Month</th>
            <th class="table-header-vertical">Q1</th>
            <th class="table-header-vertical">Q2</th>
        </tr>
    </thead>
</table>

<!-- Book spine -->
<div class="book-spine">CSS Complete Guide</div>
```

Chapter 5

Advanced Techniques

5.1 Responsive Vertical Text

```
/* Desktop - vertical layout */
@media (min-width: 1024px) {
    .content {
        writing-mode: vertical-rl;
        text-orientation: mixed;
        columns: 2;
        column-gap: 2rem;
        height: 600px;
    }
}

/* Tablet - mixed layout */
@media (max-width: 1023px) and (min-width: 768px) {
    .content {
        writing-mode: horizontal-tb;
        text-orientation: mixed;
        columns: 2;
    }
}

/* Mobile - horizontal layout */
@media (max-width: 767px) {
    .content {
        writing-mode: horizontal-tb;
        text-orientation: mixed;
        columns: 1;
    }
}
```

5.2 Combined with Transforms

```
/* Vertical text with rotation */
.rotated-vertical {
    writing-mode: vertical-rl;
```

```
    text-orientation: mixed;
    transform: rotate(180deg);
}

/* Flipped vertical text */
.flipped-vertical {
    writing-mode: vertical-rl;
    text-orientation: mixed;
    transform: scaleX(-1);
}

/* Skewed vertical text */
.skewed-vertical {
    writing-mode: vertical-rl;
    text-orientation: upright;
    transform: skew(10deg);
}
```

5.3 Animation with Text Orientation

```
/* Fade in vertical text */
@keyframes fadeInVertical {
    from {
        opacity: 0;
        transform: translateX(20px);
    }
    to {
        opacity: 1;
        transform: translateX(0);
    }
}

.animated-vertical {
    writing-mode: vertical-rl;
    text-orientation: mixed;
    animation: fadeInVertical 0.6s ease-out;
}

/* Slide in sideways */
@keyframes slideInSideways {
    from {
        opacity: 0;
        transform: translateY(20px);
    }
    to {
        opacity: 1;
        transform: translateY(0);
    }
}

.sideways-animated {
    writing-mode: sideways-rl;
}
```

```
    text-orientation: sideways;
    animation: slideInSideways 0.8s ease-out;
}
\end{austlisting}

\newpage

\chapter{Accessibility and Best Practices}

\section{Accessibility Considerations}

\begin{lstlisting}[language=HTML]
<style>
  /* Ensure sufficient contrast */
  .vertical-text {
    writing-mode: vertical-rl;
    text-orientation: mixed;
    color: #333;
    background: white;
  }

  /* Focus visible for keyboard navigation */
  .vertical-link:focus {
    outline: 2px solid #0066cc;
    outline-offset: 2px;
  }

  /* Support dark mode */
  @media (prefers-color-scheme: dark) {
    .vertical-text {
      color: #e0e0e0;
      background: #1a1a1a;
    }
  }

  /* Respect reduced motion */
  @media (prefers-reduced-motion: reduce) {
    .animated-vertical {
      animation: none;
    }
  }
</style>

<!-- Provide lang attribute for proper speech synthesis -->
<article class="vertical-text" lang="ja">

</article>

<article class="vertical-text" lang="zh">

</article>
```

5.4 Best Practices

1. **Use proper lang attributes:** Always specify language for screen readers
2. **Test readability:** Ensure vertical text is still easily readable
3. **Provide fallbacks:** Not all browsers support vertical text equally
4. **Mobile considerations:** Vertical text may not work well on small screens
5. **Font selection:** Use fonts that support CJK characters
6. **Alternative text:** Provide horizontal alternative if needed
7. **Performance:** Limit vertical text to essential content

5.5 Font Selection

```
/* Japanese fonts */
.japanese {
    font-family: 'Noto Sans JP', 'Hiragino Kaku Gothic Pro',
                'MS Pgothic', sans-serif;
}

/* Chinese fonts */
.chinese {
    font-family: 'Noto Sans CJK SC', 'SimSun',
                'Microsoft YaHei', sans-serif;
}

/* Korean fonts */
.korean {
    font-family: 'Noto Sans CJK KR', 'Noto Serif CJK KR',
                'Malgun Gothic', sans-serif;
}

/* Fallback for all CJK */
.cjk {
    font-family: 'Noto Sans CJK JP', 'Noto Sans CJK SC',
                'Noto Sans CJK KR', sans-serif;
}
```


Chapter 6

Real-World Examples

6.1 Example 1: Vertical Magazine Layout

```
<style>
  .magazine {
    display: grid;
    grid-template-columns: 200px 1fr;
    gap: 2rem;
    padding: 2rem;
    max-width: 1200px;
  }

  .magazine-sidebar {
    writing-mode: vertical-rl;
    text-orientation: mixed;
    border-right: 3px solid #0066cc;
    padding-right: 1.5rem;
    font-weight: 600;
    display: flex;
    align-items: center;
  }

  .magazine-content {
    writing-mode: vertical-rl;
    text-orientation: mixed;
    columns: 3;
    column-gap: 1.5rem;
    column-rule: 1px solid #ddd;
  }

  .magazine-content h2 {
    column-span: all;
    writing-mode: horizontal-tb;
    margin-bottom: 1rem;
  }
</style>

<div class="magazine">
  <aside class="magazine-sidebar"> </aside>
```

```

<article class="magazine-content">
  <h2>Article Title</h2>
  <p>Magazine content in vertical layout...</p>
</article>
</div>

```

6.2 Example 2: Data Table with Vertical Headers

```

<style>
  .vertical-table {
    border-collapse: collapse;
    margin: 2rem 0;
  }

  .vertical-table thead th {
    writing-mode: vertical-rl;
    text-orientation: upright;
    padding: 1rem 0.5rem;
    background: #f5f5f5;
    border: 1px solid #ddd;
    height: 200px;
    font-weight: 600;
  }

  .vertical-table tbody td {
    padding: 1rem;
    border: 1px solid #ddd;
    text-align: center;
  }

  .vertical-table tbody tr:nth-child(even) {
    background: #f9f9f9;
  }
</style>

<table class="vertical-table">
  <thead>
    <tr>
      <th>January</th>
      <th>February</th>
      <th>March</th>
      <th>April</th>
    </tr>
  </thead>
  <tbody>
    <tr>
      <td>1,200</td>
      <td>1,500</td>
      <td>1,800</td>
      <td>2,100</td>
    </tr>
  </tbody>
</table>

```

```
</table>
```

Chapter 7

Quick Reference

7.1 text-orientation Values

```
/* Mixed - default for vertical-rl, vertical-lr */
text-orientation: mixed;

/* Upright - all characters upright */
text-orientation: upright;

/* Sideways - all characters rotated 90 degrees */
text-orientation: sideways;

/* Use font's natural orientation */
text-orientation: use-glyph-orientation;

/* Global values */
text-orientation: inherit;
text-orientation: initial;
```

7.2 Common Combinations

```
/* Japanese vertical - most common */
.japanese-vert {
    writing-mode: vertical-rl;
    text-orientation: mixed;
}

/* Chinese vertical */
.chinese-vert {
    writing-mode: vertical-lr;
    text-orientation: upright;
}

/* Sideways text */
.sideways {
    writing-mode: sideways-rl;
    text-orientation: sideways;
```

```
}

/* Chart/graph labels */
.chart-label {
    writing-mode: vertical-rl;
    text-orientation: upright;
    transform: rotate(180deg);
}

/* Book spine */
.book-spine {
    writing-mode: vertical-rl;
    text-orientation: mixed;
}
```

7.3 Production Template

```
/* Complete text-orientation setup */

/* Base vertical styles */
.vertical-text {
    writing-mode: vertical-rl;
    text-orientation: mixed;
    line-height: 2;
}

/* Language-specific */
.japanese {
    font-family: 'Noto Sans JP', sans-serif;
    lang: jp;
}

.chinese {
    font-family: 'Noto Sans CJK SC', sans-serif;
    lang: zh;
}

/* Responsive */
@media (max-width: 768px) {
    .vertical-text {
        writing-mode: horizontal-tb;
        text-orientation: mixed;
    }
}

/* Accessibility */
@media (prefers-reduced-motion: reduce) {
    .vertical-text {
        animation: none;
    }
}
```

```
/* Dark mode */
@media (prefers-color-scheme: dark) {
  .vertical-text {
    color: #e0e0e0;
  }
}
```